DETAILED PROJECT REPORT

Title

House Sales Price Prediction Using Machine Learning
(King County, USA Dataset)

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1. Introduction

Real estate price prediction plays a vital role for buyers, sellers, and investors. Accurate price estimation helps in informed decision-making and market analysis.
This project focuses on predicting house prices in **King County, USA**, using historical housing data and machine learning techniques.

The project includes:

- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Data visualization
- Building a regression model to predict house prices

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2. Dataset Description

The dataset contains **21,613 house sale records** with **21 attributes**.

Key Features

Feature	Description
price	House sale price (Target variable)
bedrooms	Number of bedrooms
bathrooms	Number of bathrooms
sqft_living	Living area in square feet
sqft_lot	Lot size in square feet
floors	Number of floors
waterfront	Waterfront view (0/1)
condition	Condition of the house
grade	Overall grade

Feature	Description
sqft_above	Area above basement
sqft_basement	Basement area
yr_built	Year built
yr_renovated	Year renovated
zipcode	Location
lat, long	Geographical coordinates

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3. Tools and Libraries Used

- **Python**
- **NumPy** – Numerical operations
- **Pandas** – Data handling
- **Matplotlib & Seaborn** – Visualization
- **Scikit-learn** – Machine learning model

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4. Data Loading and Inspection

- Dataset loaded using pandas.read_csv()
- Checked:
    - Shape of dataset
    - Column names and data types
    - Missing values
    - Duplicate entries
- Initial observation showed that the dataset is mostly clean with minimal preprocessing needed.

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5. Data Cleaning and Preprocessing

Steps performed:

- Converted date column into datetime format
- Checked and confirmed no significant missing values
- Removed irrelevant columns (like id)

- Ensured numerical consistency
- Target variable: **price**
- Feature scaling applied where required

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6. Feature Engineering

- Extracted **year** from date
- Removed outliers from price and sqft_living
- Created meaningful relationships between variables

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7. Exploratory Data Analysis (EDA)

Key Findings

- House prices are **right-skewed**
- Higher prices are associated with:
  - Larger living area
  - Better house grade
  - Waterfront properties
- sqft_living shows strong positive correlation with price
- bedrooms alone is not a strong predictor

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8. Data Visualizations

Visualizations used:

- Histogram of house prices
- Correlation heatmap
- Scatter plot of sqft_living vs price
- Boxplot for bedrooms vs price
- Price distribution by waterfront

These plots helped identify trends and relationships clearly.

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9. Machine Learning Model

Model Used

- **Linear Regression**

Steps

1. Split dataset into training and testing sets (80:20)
2. Trained Linear Regression model
3. Predicted house prices
4. Evaluated performance

Evaluation Metrics

- **R^2 Score**
- **Mean Absolute Error (MAE)**
- **Mean Squared Error (MSE)**

The model showed reasonable accuracy, indicating that linear relationships exist between price and features.

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10. Results and Discussion

- Model successfully learned price trends
- Living area, grade, and location significantly impact price
- Prediction accuracy can be improved using advanced models

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11. Conclusion

This project demonstrated how machine learning can be applied to real-world housing data to predict prices. The Linear Regression model provided meaningful insights, though more complex models can further improve accuracy.

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12. Future Enhancements

- Use advanced models (Random Forest, XGBoost)
- Feature selection and hyperparameter tuning
- Location-based clustering
- Deployment as a web application